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KCDC 2026 · KANSAS CITY, MO · SEPT 10–11

When Logs Tell the Whole Story

OpenTelemetry for Beginners

Barret Blake · Architect, NimblePros

#KCDC26 | #OTel | #Observability

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WHO AM I

Barret Blake

Barret Blake

Architect · NimblePros

Microsoft® Most Valuable Professional

A few things about me

- Microsoft MVP — .NET
- Husband & Father
- Coder, Blogger & Speaker
- Mentor
- Gamer
- Model Railroader & LEGO Enthusiast
- Buckeyes & Browns Fan

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THE PROBLEM

Production failures are inevitable

What happens next is up to you.

No observability

Guessing in the dark. Each incident is a mystery. MTTR spirals.

Scattered logs

Logs exist but tell disconnected stories. Hard to correlate across services.

Structured telemetry

A clear, connected story. Logs trace the path from failure to root cause.

The difference between a quick fix and a prolonged outage **often comes down to whether your logs tell a clear, connected story.**

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FOUNDATION

What is OpenTelemetry?

An open-source, language-agnostic, vendor-neutral framework for application observability — providing APIs, SDKs, and tools to capture traces, metrics, and logs.

Open Source

Backed by the CNCF. Community driven and widely adopted.

Language Agnostic

.NET, Java, Go, JS, Python, Rust, Ruby & more.

Vendor Neutral

Send your data to ANY compatible observability platform.

The industry standard for cloud-native observability.

CORE CONCEPTS

The Three Signals of OTel

Traces

The full path of a request through your system — entry point to database and back.

Metrics

Numeric measurements over time — request counts, error rates, latency percentiles.

Logs

Timestamped records of what happened and why. The base unit of all telemetry.

Baggage — not a signal, but cross-cutting context: key-value data propagated across all spans and service boundaries within a trace.

Coming soon: Profiles — an emerging fourth signal (public alpha, 2026) answering “which code is burning CPU / memory?”

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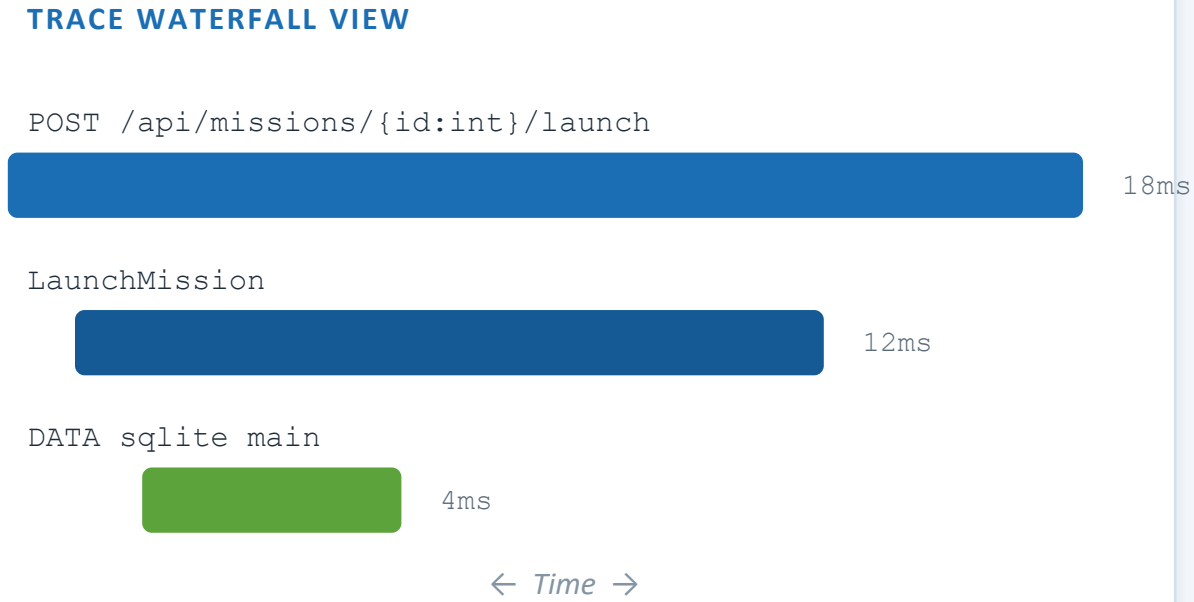
TRACES

Traces: Following the Request

A trace records the complete path of a request through your application.

- Each unit of work = a Span
- Spans nest into parent / child trees
- Captures timing for every step
- Spans across services share a TraceId
- Critical for cross-service debugging



Real-world impact: Analyzing trace waterfalls on a client project revealed several performance bottlenecks — a **10–40% runtime reduction** across key workflows.

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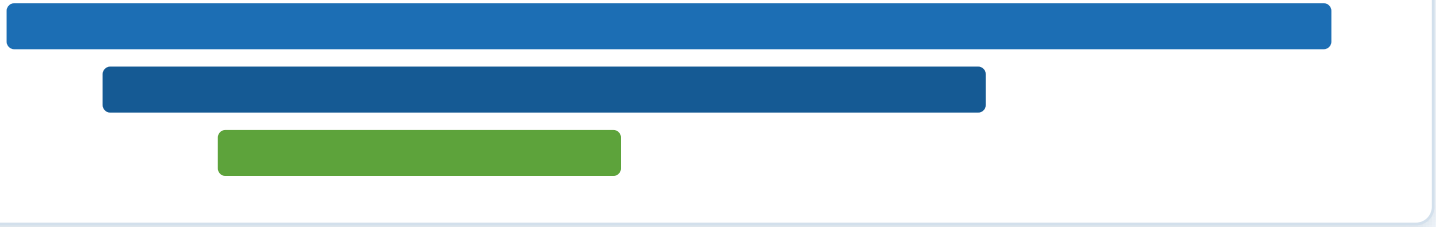
LOGS + TRACES

Trace Context: Connecting the Dots

The same TraceId and SpanId appear in both your logs AND your trace spans.

LOG ENTRY

```
TraceId:
26bf82c5...
```



One click to context

Jump from a log line directly to the full trace waterfall in your platform.

Cross-service correlation

The same TraceId flows through all services — find a user's request across microservices instantly.

No manual correlation

OTel propagates trace context automatically via HTTP headers and messaging metadata.

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METRICS

Metrics: Measuring What Matters

Error-Rate Alerting

Don't alert on every failure — you'll train your team to ignore alerts. Use a metric counter and fire only when a threshold is crossed (e.g. 20+ failures in 60 seconds).

Code Usage Analysis

Before investing in a refactor, use metrics to prove it's worth it. If a 1,000-line class runs hundreds of times per minute, that data is your business case.

```
// Custom counter setup
public MissionSimulator(IMeterFactory meterFactory)
{
    var meter =
meterFactory.Create(MissionTelemetry.MeterName);

    _warp = meter.CreateGauge<double>("mission.warp_speed",
unit: "warp", description: "Current warp factor of an in-
flight mission.");
    _warp.Record(run.WarpSpeed, shipTag);

    _launches =
meter.CreateCounter<long>("missions_launched", unit:
"{launch}", description: "Missions launched, by ship.");
    _launches.Add(1, shipTag);
}
```

Metric output: 3 launches for Enterprise, 1 for Voyager.

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LOGS

Logs: The Story Behind the Numbers

> ***"Not all logs are events, but all events are logs."*** — OpenTelemetry docs

Automatically shipped

OTel exports the log records you already create — the exporter handles it, no special wiring.

Structured output

OTel turns any log format into a structured record: timestamp, severity, body, attributes.

Correlated with traces

Every log within a trace carries the TraceId and SpanId — connecting logs to the request.

When you need to understand what went wrong and why — logs are where you look.

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LOGS

Anatomy of an OTel Log Record

```
LogRecord.Timestamp:    2026-09-10T09:14…
LogRecord.TraceId:      26bf82c5…
LogRecord.SpanId:       c0da8620…
LogRecord.TraceFlags:   Recorded
LogRecord.CategoryName: Program
LogRecord.Severity:     Info
LogRecord.Body:         Mission launch
  requested: {MissionName}…
LogRecord.Attributes:
    MissionName: Enterprise
    Status: In Flight
```

Field	Purpose
`Timestamp`	When the event occurred
`TraceId`	Links log to its trace
`SpanId`	Specific span within the trace
`TraceFlags`	W3C standard trace flags
`CategoryName`	.NET ILogger category
`Severity`	Log level (Info, Warn, Error…)
`Body`	The log message
`Attributes`	Structured key-value data

```
logger.LogInformation("Mission launch requested: {MissionName}, status: {Status}", mission.Name, mission.Status);
```

CORE CONCEPTS

Baggage: Context That Travels

Key-value context that propagates automatically across all spans, services, and HTTP headers within a trace.

```
// Set baggage at request start
Baggage.SetBaggage(
    "mission.commander", "Picard");
Baggage.SetBaggage(
    "mission.priority", "High");

// Read anywhere in the trace
var commander = Baggage
    .GetBaggage("mission.commander");
```

Good uses for Baggage

- User ID for correlation
- Tenant ID in multi-tenant apps
- Source IP address
- Feature-flag or A/B testing group
- Request source / channel

Security warning: Baggage is added to HTTP headers and is readable by any receiving service. **Never put PII, passwords, or sensitive data in baggage.**

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BEST PRACTICES

Choosing the Right Log Level

Level	Use When…	Recommended For
Trace	Highly detailed debugging. Very high volume. Dev only.	Never in production
Debug	Diagnostic details for development and debugging.	Dev / staging only
Information	Normal operations. Application milestones, key events.	Production — selectively
Warning	Unexpected but recoverable. Worth monitoring.	Always in production
Error	Operation failed. Needs attention; app still running.	Always in production
Critical	Catastrophic failure. Immediate action required.	Always + alerting

ANTI-PATTERNS

Common Logging Mistakes to Avoid

Over-Logging

Problem: Logging everything buries the signal in noise — and drives up cost on commercial platforms.

Fix: Log meaningful events at appropriate levels. Filter dev-only logs from production.

Unstructured Messages

Problem: String concatenation ("Error for user " + id) can't be queried, filtered, or aggregated.

Fix: Use structured logging: LogError("User {UserId} failed checkout", id) — queryable attributes.

Missing Context

Problem: “Operation failed” is useless without which operation, which user, and what data.

Fix: Always include identifiers: user IDs, record IDs, operation names. Failures are data-specific.

Alert-Fatigue Loops

Problem: Alerting on every failure trains your team to build rules that ignore alerts entirely.

Fix: Use metric thresholds, not per-event alerts. Alert on patterns, not occurrences.

GETTING STARTED

Setting Up OTel in .NET

Step 1 — Add NuGet Packages

- `OpenTelemetry.Extensions.Hosting`
- `OpenTelemetry.Instrumentation.AspNetCore`
- `OpenTelemetry.Exporter.OpenTelemetryProtocol`

That's it for basic setup!

OTel auto-instruments your ASP.NET Core routes and starts collecting logs, traces, and metrics immediately. For Aspire projects it's included in the ServiceDefaults template — no manual setup.

```
// Step 2 — configure in Program.cs
builder.Logging.AddOpenTelemetry(o => o
    .SetResourceBuilder(ResourceBuilder
        .CreateDefault()
        .AddService("app-logger"))
    .AddOtlpExporter());

builder.Services.AddOpenTelemetry()
    .ConfigureResource(r => r
        .AddService("app-logger"))
    .WithTracing(t => t
        .AddAspNetCoreInstrumentation()
        .AddOtlpExporter())
    .WithMetrics(m => m.AddOtlpExporter());
```

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PRODUCTION

Exporting to Production Platforms

Self-Hosted

Jaeger — Distributed tracing

Grafana + Loki + Prometheus — Metrics, logs & dashboards

Elastic Stack — Search-first observability

Pros: control costs, data ownership. **Cons:** you own maintenance.

Commercial SaaS

Datadog — Full-stack observability

Grafana Cloud — Hosted Grafana + alerting

Azure App Insights — Native .NET + Azure

New Relic — AI-powered incident response

Pros: managed, enterprise features. **Cons:** cost scales with data.

The OTel advantage: portability

Everything speaks **OTLP**, the OpenTelemetry wire protocol. Route through the **OpenTelemetry Collector** and switching platforms is a config change — not a rewrite. Your data model stays the same.

TOOLING

Aspire: OTel Built In for Local Dev

Console View

Raw log output across all services in one pane.

Structured Logs

Filter and search by severity, category, or attribute.

Trace List

Browse all traces, filter by service or duration.

Waterfall View

Visual span timing — pinpoint slow code instantly.

Metrics

Runtime metrics and custom counters in one dashboard.

Multi-Language

Aspire 13 is now a multi-language platform — .NET, Python & JS/Node.

Don't deploy the Aspire dashboard to production. It exposes full debug logs with no auth. Use a dedicated observability platform in deployed environments.

ADVANCED PATTERNS

Reusable Telemetry with ServiceDefaults

The ServiceDefaults shared project is the perfect home for reusable telemetry across your whole solution.

Custom Metric Counters

Define counters in ServiceDefaults via an AddMissionMetrics extension, then call counter.Add(1) in any endpoint. Metrics appear instantly in the dashboard.

Reusable Span Factory

A generic StartDatabaseSpan<T>() auto-tags span attributes from your entity's properties. One method, consistent tracing everywhere.

```
// ServiceDefaults: reusable span factory
public static Activity? StartDatabaseSpan<T>(
    this ActivitySource src, string name, T entity)
{
    var activity = src.StartActivity(
        name, ActivityKind.Internal);
    if (activity is null) return activity;
    // auto-tag entity properties…
    foreach (var p in typeof(T).GetProperties())
        activity.SetTag($"db.entity.{p.Name}",
                        p.GetValue(entity));
    return activity;
}
```

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DEMO TIME

Mission Control

Launch a mission → watch traces, metrics, logs & baggage light up the Aspire dashboard in real time.

Traces | Metrics | Logs | Baggage

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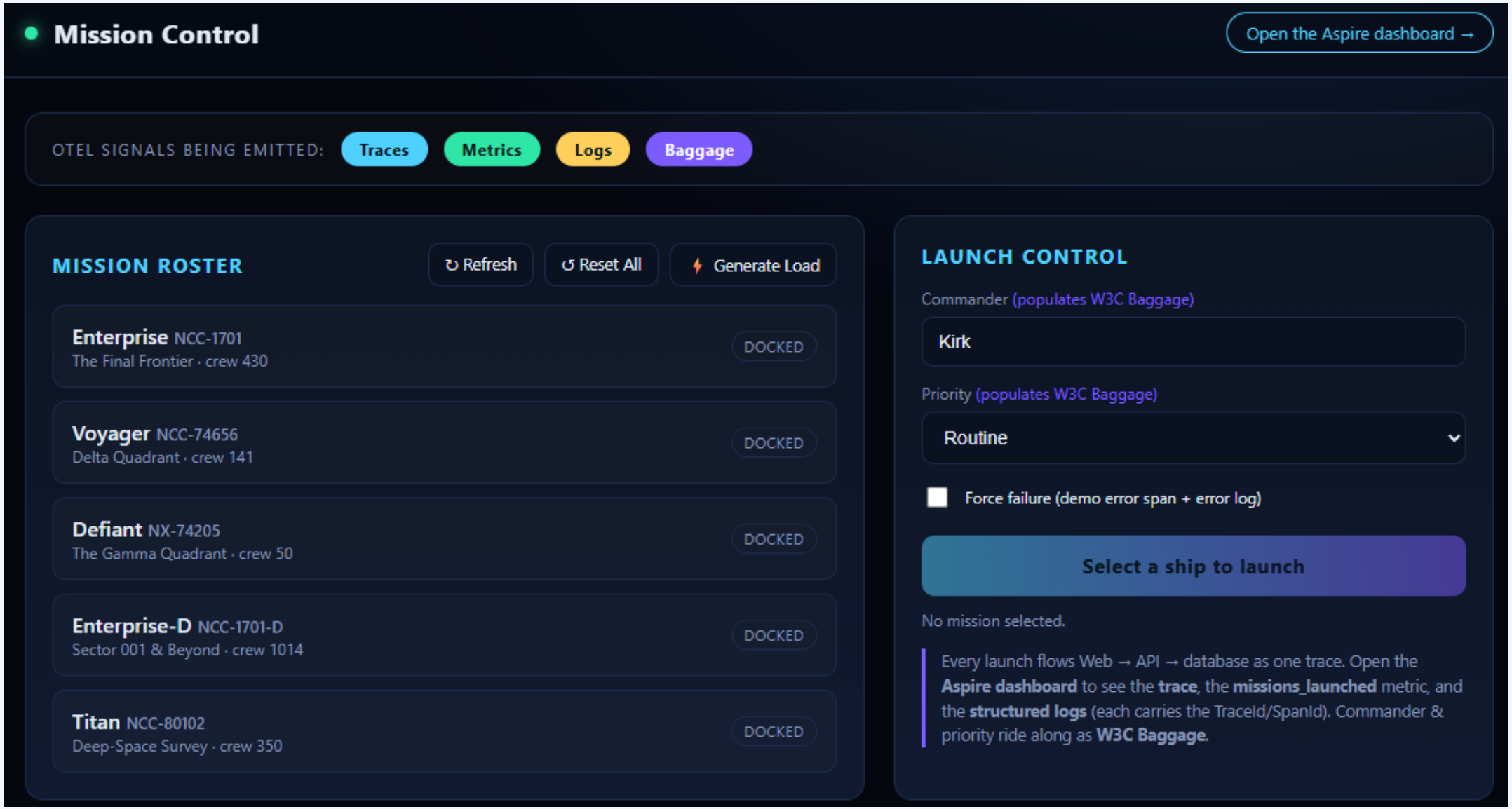
BACKUP

Mission Control — the app



After launch, warp / shields / torpedoes stream live every 2s into traces and metrics.

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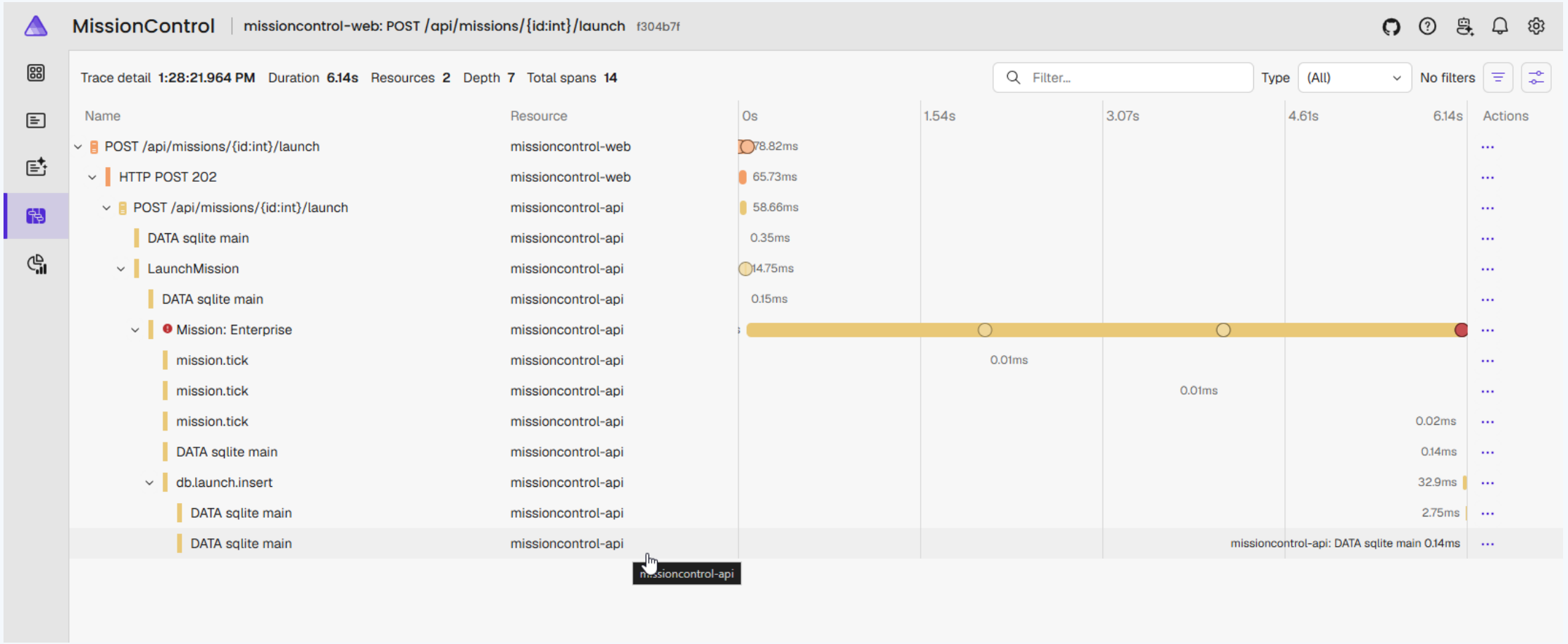
BACKUP

Aspire Dashboard — Trace Waterfall



*One long-lived Mission span; a **mission.tick** child every 2s captures warp, shields & torpedoes. Commander rides along as Baggage.*

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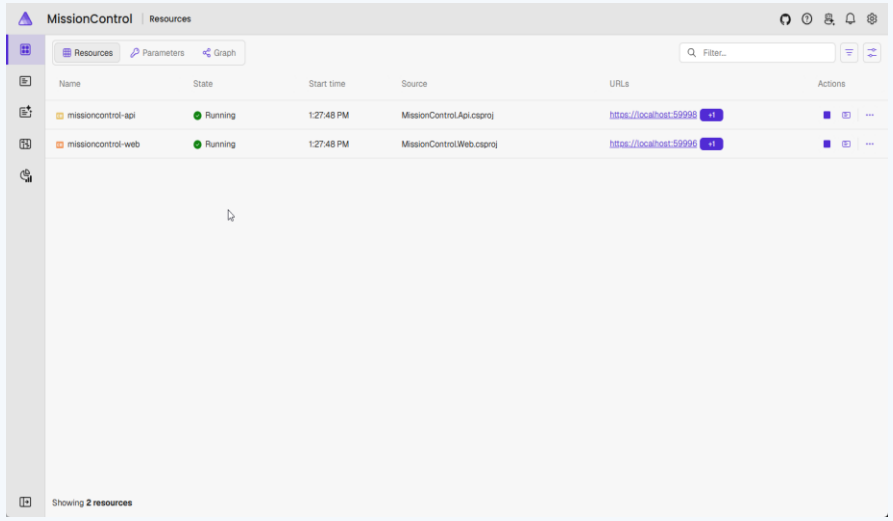
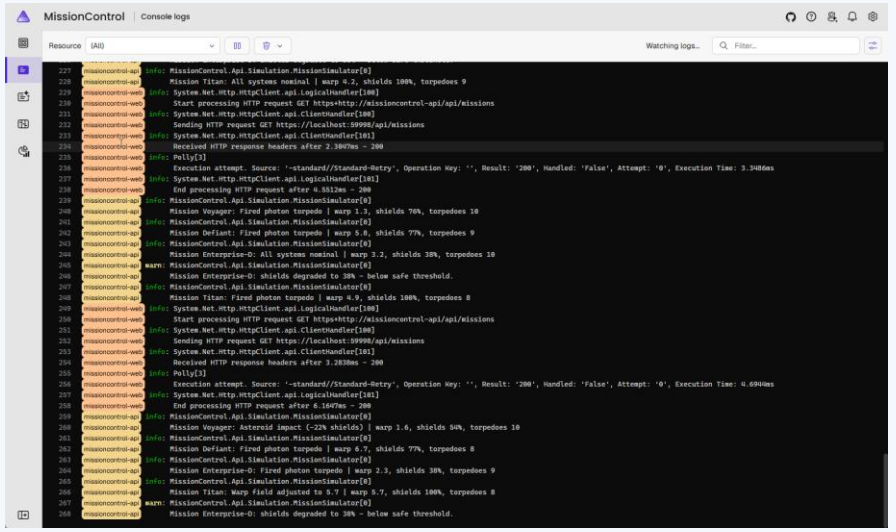
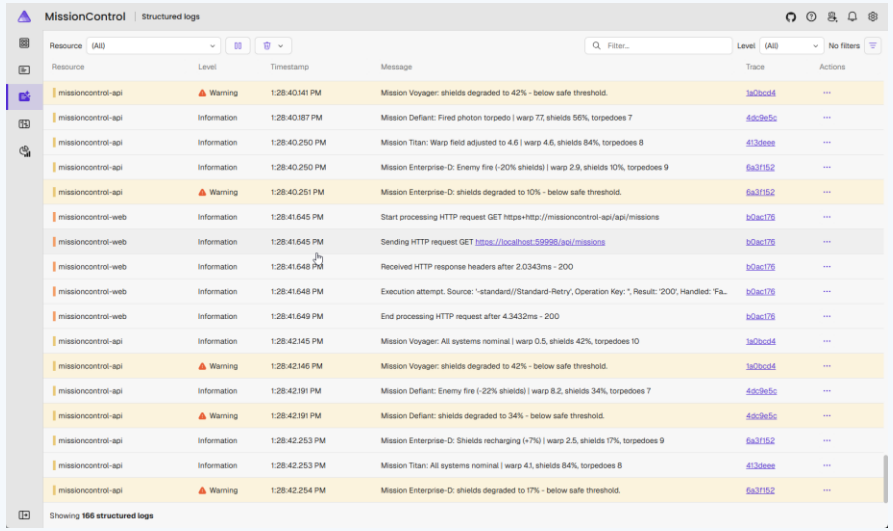
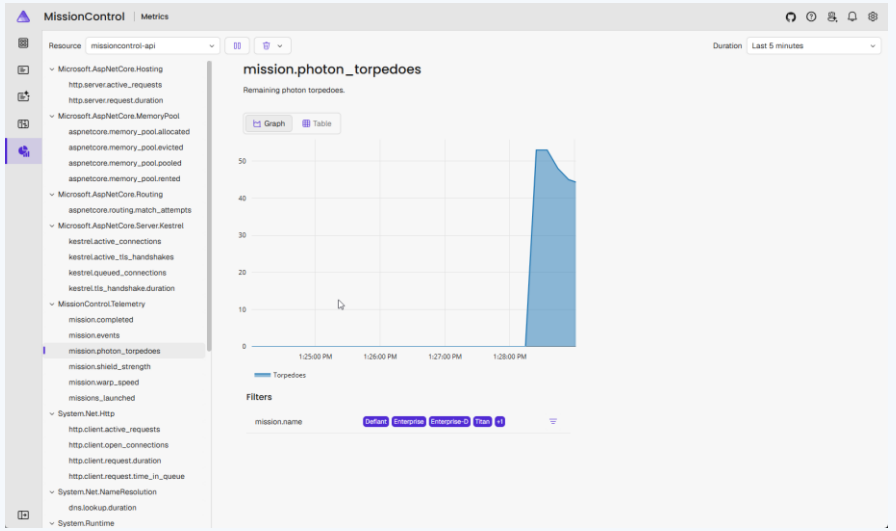
BACKUP

Aspire Dashboard — Metrics & Structured Logs









The same TraceId *runs through every 2s tick — tying the gauges, the logs, and the Mission trace together.*

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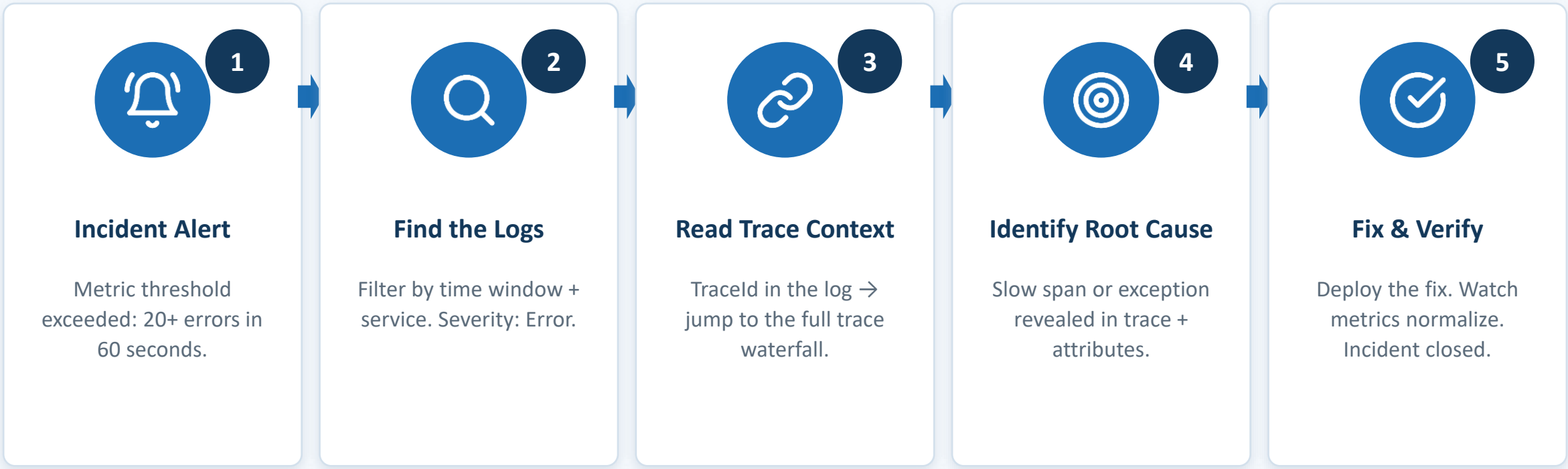
SUMMARY

The Observability Stack in Action



WRAP-UP

Key Takeaways

OpenTelemetry is the industry standard — vendor-neutral, language-agnostic observability for modern apps.

Logs + Traces + Metrics work together. The TraceId connects log entries to full request paths across services.

Log-level discipline matters. Use structured messages with meaningful attributes — not noisy string concatenation.

Baggage propagates context automatically — great for tenant ID, user ID, feature flags. Never PII.

Avoid over-logging and alert fatigue. Alert on patterns with metric thresholds, not individual events.

Aspire ServiceDefaults makes telemetry reusable. Define once, instrument everywhere — spans and counters.

OTel gives you freedom to switch platforms. Your investment is portable — a config change, not a rewrite.

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Thank You, Kansas City!

Questions & Discussion

Barret Blake · Architect, NimblePros

Session Feedback

Resources

- opentelemetry.io/docs/what-is-opentelemetry/
- opentelemetry.io/docs/languages/dotnet/
- aspire.dev/dashboard/
- learn.microsoft.com — .NET observability with OTel
- github.com/barretb/MissionControlDemo

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